

# Vipin Kumar Karthikeyan

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## Experience

### Technical Business Intelligence Analyst

#### Waters Corporation - Milford, MA

February 2026 – Present

- Partnered with business stakeholders to analyze and contribute to a Streamlit-based enterprise analytics application hosted on Databricks, supporting data transformation workflows across medallion layers using Python and Unity Catalog.
- Developed and maintained Databricks notebooks and data pipelines integrating enterprise sources into centralized analytics environments, delivering BI reporting solutions aligned to organizational KPIs and data governance standards.

### Cloud Analytics Intern

#### Waters Corporation - Milford, MA

June 2025 – December 2025

- Architected multi-cloud cost intelligence platform across 45+ Azure and 12 AWS accounts using Lambda, Data Factory, and Power BI, processing cost data from APIs into centralized storage with automated daily refresh.
- Engineered ServiceNow integration with Python automation for incident management and ticket routing, implementing Jira workflows for cross-functional collaboration, reducing manual effort by 85%.
- Implemented real-time alerting with threshold-based monitoring, service principal authentication, and automated email notifications for cost anomalies and budget variances, delivering actionable insights for optimization.

### Data & Business Intelligence Engineer

#### Thorogood Associates - Bangalore, India

July 2022 - July 2024

- Architected end-to-end medallion data lakehouse using Azure Data Factory, Databricks clusters, and Microsoft SQL Server, processing 500K+ invoices with CI/CD automation achieving 99.9% data accuracy.
- Designed and implemented a cloud-based sustainability analytics platform, integrating SAP ERP systems and factory-level data into Azure Data Lake, boosting sustainable procurement visibility by 34%.
- Streamlined batch processing workflows using Databricks SQL, MLflow, and Power Automate, reducing processing time by 30% with automated data validation through Power Apps.
- Built and optimized enterprise BI dashboards using Power BI and Azure Analysis Services, supporting \$3.06B in sales analysis and identifying \$65.2M in lost sales across 15+ categories.
- Streamlined data processing and reporting by implementing partitioned delta tables, improving Power BI refresh performance by 40% and enhancing time-to-insight for critical business users.

### Data Engineer Intern

#### EXL Service - Chennai, India

December 2021 - June 2022

- Migrated on-premise SQL Server infrastructure to cloud-native architectures using AWS (S3, Lambda, Kinesis, Aurora) and Azure, reducing compute costs by 35% and improving query performance by 60%.
- Developed scalable ETL pipelines using AWS Glue and Azure Data Factory, processing 500K+ medical claims per file, enabling real-time ingestion and 350K+ edits/sec in fraud detection scenarios.
- Designed end-to-end data orchestration with Databricks, Service Bus, and Azure Blob Storage, ensuring secure audit trails and SLA-compliant pre-pay and post-pay claim analytics.

### Data Scientist Intern

#### Neewee Analytics - Bangalore, India

May 2020 - July 2021

- Deployed a predictive maintenance solution for steam turbines using Multivariate State Estimation Technique (MSET) and Statistical Process Control, reducing unplanned downtime by 25%.
- Automated fault detection workflows with data imputation, feature engineering, and dashboarding, cutting manual monitoring efforts and lowering maintenance costs by 20%.

## Academic Projects

### Voice-to-Text Q&A Chatbot

September 2024 – Present

- Architected an Agentic AI chatbot on AWS, integrating Lambda, Transcribe, and S3 for speech-to-text processing and dynamic task delegation based on intent classification and user prompts.
- Deployed Retrieval-Augmented Generation (RAG) pipelines with Amazon Bedrock and SageMaker, enabling real-time interactions with LLM-based responses fine-tuned for domain-specific Q&A tasks.

## Skills

- Programming Languages:** Python, SQL, PySpark, Scala, Java, C, C++, NoSQL, R
- Cloud Platforms:** Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), Snowflake
- Data Engineering:** Airflow, Kafka, Data Warehousing, Relational databases, Stream Processing, Data Lakes & Lakehouses
- Data & Analytics:** Data Modeling, Data Transformation, AI, Data Architecture, Data Quality & Governance, Statistical Analysis
- Visualization & BI Tools:** Microsoft Power BI, Tableau, Excel, DAX, M Query, QlikView, Looker
- Core Competencies:** Project Management, Agile Methodologies, Stakeholder Engagement, Cross-functional Collaboration

## Education

### University of Massachusetts Amherst - Isenberg School of Management

Master of Science in Business Analytics

September 2024 - December 2025

### Anna University - Coimbatore Institute of Technology

Master of Science in Data Science

July 2017 - May 2022

## Certifications

- Azure Certified Data Engineer Associate - DP-203: Data Engineering on Microsoft Azure
- Databricks Certified Data Engineer Associate